The 2024 and 2025 Stack Overflow Developer Survey datasets are relational. Each response maps to a single respondent across a consistent schema. Fields such as OrgSize, DevType, and Country carry well-defined categorical values with little variance between survey years. This consistency favours SQL. In practice, survey data of this kind is routinely stored in PostgreSQL because schema predictability makes query optimisation straightforward and cross-year comparisons require little more than a UNION or an additional year column. The analytical operations performed in this project, frequency counts and grouped comparisons by role and organisation size, are exactly the workloads for which relational databases were created.

One column complicates this. CICDTools stores multiple tool selections as a semicolon-delimited string, violating first normal form and requiring string splitting before any analysis can begin. This pattern is common in enterprise data exports, where simplicity of ingestion takes priority over downstream usability. A document-oriented store such as MongoDB would represent each response as a nested document, preserving multi-value selections natively and eliminating the need for that pre-processing step.

SQL remains the stronger choice for this dataset. The CICDTools anomaly is the exception, not the rule. Introducing NoSQL flexibility across two years of otherwise well-structured data would create more complexity than it resolves.